

Ali Ahmadi Esfidi

AI Researcher & Developer

✉ aliahmadiesfidi@outlook.com | 🌐 Mr-Ahmadi | 📄 Ali Ahmadi Esfidi | 🌐 Portfolio

Education

- B.Sc. Computer Science** > Amirkabir University of Technology 📅 Sep 2022 – Present
● Location: Tehran, Iran | Field of study: Computer Science | GPA: 18.90/20 | Top 20%
- High School Diploma** > Dr. Shahriari High School (SAMPAD) 📅 Sep 2019 – Jun 2022
● Location: Qom, Iran | Field of study: Mathematics & Physics | GPA: 18.42/20

Research Interests

- Deep Reinforcement Learning
- Machine Learning for Decision-Making Systems
- AI for Infrastructure Monitoring
- Optimization Methods for AI and Networking
- Computational Biology & Bioinformatics
- Software-Defined Networks & Resource Allocation

Experiences

- Research Assistant** > Supervisor: Prof. Ghatte 📅 Jan 2025 – Present
📍 NORC Lab, Amirkabir University of Technology
- **Led** an AI team and **engineered** web systems and deep learning models for automated bridge damage detection in collaboration with the Tehran Urban Planning and Research Center.
 - **Gained hands-on experience** in developing efficient deep-learning pipelines for computer vision applications, focusing on model compactness and structural health monitoring.
 - *Stack::* **PyTorch** • **OpenCV** • **ReactJS** • **NodeJS** • **PostgreSQL** • **Docker**
- Research Assistant** > Supervisor: Prof. Khonsari 📅 Sep 2024 – Present
📍 High Performance Networks Lab, Tehran University
- **Developed** deep reinforcement learning systems for complex agricultural decision-making, focusing on reward engineering and policy optimization to balance multiple objectives.
 - **Acquired proficiency** in benchmarking reinforcement learning algorithms across discrete and continuous action spaces, and in utilizing simulation environments for training and evaluation.
 - *Stack::* **PyTorch** • **TensorFlow** • **Pandas** • **NumPy**
- Research Assistant** > Supervisor: Dr. Dolati 📅 Sep 2025 – Present
📍 Sharif University of Technology
- **Formulated** MDPs for switch pairing and collaborative rule caching, and designed simulation environments for joint scheduling-routing policies in programmable data planes.
 - **Implemented** optimization frameworks for in-network aggregation in distributed machine learning, focusing on ILP decomposition, flexible routing strategies, and scalable algorithmic design.
 - *Stack::* **PyTorch** • **Numba** • **Pandas** • **NumPy**
- Teaching Assistant** > Graduate & Undergraduate Courses 📅 Sep 2024 – Present
📍 Amirkabir University of Technology & Tehran University
- **Quantum Information Processing** (Master's, Prof. Khonsari) — Sep 2025 – Jan 2026
 - **Computational Data Mining** (Master's, Prof. Ghatte) — Sep 2025 – Jan 2026
 - **Theory of Computation** (Dr. Didehvar) — Sep 2024 – Jun 2026
 - **Introduction to Logic** (Dr. Didehvar) — Sep 2025 – Jan 2026
 - **Artificial Intelligence & Workshop** (Prof. Ghatte, Dr. Yousefimehr) — Sep 2024 – Jan 2025

Publications

- [1] P. Heidari, A. Ahmadi Esfidi, A. Mehrvarz, E. Khodaei, A. Khonsari, **Irrigation Optimization in Agricultural Fields Using Deep Reinforcement Learning Approaches**, Published in *CSICC 2025*. [View Paper](#).
- [2] P. Heidari, A. Khonsari, A. Ahmadi Esfidi, A. Mehrvarz, E. Khodaei, A. Dadlani, **DeepIrrigo: Deep Reinforcement Learning for Continuous and Discrete Action Irrigation Optimization — A Case Study in Iran**, Under review at *Intelligent Systems with Applications*. [View Paper](#).
- [3] M. Saberi, A. Ahmadi Esfidi, M. Dolati, A. Khonsari, A. Movaghar, **Rule Caching in Programmable Networks with Deep Reinforcement Learning**, Submitted to *IEEE Transactions on Network and Service Man-*

agement.

[4] M. Saberi, A. Ahmadi Esfidi, M. Dolati, N. Omidvar, **FlexINA: In-Network Aggregation for Accelerating Distributed Machine Learning with Flexible Routing**, Submitted to *IEEE Transactions on Network and Service Management*.

Selected Projects

 [See more](#)

DRL-Based Beamforming Optimization for Movable Intelligent Surfaces Jun 2026 – Present

- **Designed** a DRL framework to optimize beamforming for MIS-assisted SWIPT systems, jointly configuring phase shifts and power splitting ratios to maximize energy harvesting.
- *Stack:* [PyTorch](#) • [Gymnasium](#) • [MOSEK](#) • [NumPy](#)

RL-Based ML Job Scheduler with Distribution Optimization Aug 2025 – Oct 2025

- **Developed** a two-stage hierarchical PPO system for scheduling ML jobs on multi-server accelerator clusters using the Gavel dataset, reducing average job completion time.
- *Stack:* [PyTorch](#) • [Gymnasium](#) • [NumPy](#)

Machine Learning Perspectives on Gold-Binding Peptides May 2025 – Jun 2025

- **Applied** machine learning to classify and predict gold-binding affinity of peptide sequences, comparing regression and classification approaches across multiple featurization strategies.
- *Stack:* [PyTorch](#) • [Scikit-learn](#) • [Hugging Face](#) • [Pandas](#) • [NumPy/SciPy](#)

RNA Pairing Pattern Recognition Model Mar 2024 – Jul 2024

- **Developed** a system to predict RNA secondary structures using SCFGs, evolutionary information, and enhanced CYK parsing for complex motifs; presented at [CBRC Journal Club](#).
- *Stack:* [NetworkX](#) • [Biopython](#) • [PhyML](#) • [NumPy/SciPy](#)

Skills

Programming Languages	Python • JavaScript • TypeScript • R • Java • C++ • Swift
Machine Learning Frameworks	PyTorch • TensorFlow • Pandas • Scikit-learn • NumPy • OpenCV • LangChain
Tools & Platforms	Docker • Git • GitHub • MongoDB • PostgreSQL • Hugging Face

Languages

Persian	Native
English	Fluent

Certifications

Bioinformatics Internship Program  Biocan	 Nov 2025
Scientific Secretary: Dr. Z. Salehi • Course Director: Dr. K. Kavousi	
Introduction to Bioinformatics  Biocan	 Jul 2025
Scientific Chair: Dr. K. Kavousi • Score: 96/100 • Top 3% of 350+ students	
Scrum Foundations Course  Ultima Training Tech	 Dec 2024
Instructor: Josef Balahan	

Volunteering

Student Scientific Magazine “Halghe”  Amirkabir University of Technology	 Dec 2025 – Present
Contributed to scientific editing and supported the editorial team	
Technical & Web Support Volunteer  AI Academic Events	 May 2023, Feb 2025
Web and procurement support for DAI DAY (2025) and the International Conference on AI and Smart Vehicles (2023)	